

REAPER-Notator

Complete User Manual & Functional Reference

Native Score Editing, Engraving & DAW Automation for Cockos REAPER

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Manual Contents Summary

- Ch. 1:** Introduction & Interface Architecture
- Ch. 2:** Top Toolbar & Global Navigation
- Ch. 3:** Score Canvas & Note Editing Engine
- Ch. 4:** Standard-Compliant Automatic Beaming Engine
- Ch. 5:** Dynamics Engine, Hairpins & Bow Swell Function
- Ch. 6:** Articulations & Rearticulate Integration
- Ch. 7:** Holding & Sustain Pedal Lane (CC64)
- Ch. 8:** Tempo Maps & Metric Transitions
- Ch. 9:** Octave Shift Lines (8va, 8vb, 15ma, 15mb)
- Ch. 10:** Orchestral Pattern Browser & 1-Click Library
- Ch. 11:** Rehearsal Marks, Fermatas & Score Tools
- Ch. 12:** MusicXML 4.0 Import & Export
- Ch. 13:** Score Clipboard & Selective Copy/Paste Engine
- Ch. 14:** Engine Architecture & Performance Optimization
- Ch. 15:** Keyboard Shortcuts & Quick Reference

Chapter 1: Introduction & Interface Architecture

1.1 Overview & Vision

REAPER-Notator is a native, professional music scoring, notation editing, and engraving environment built directly into Cockos REAPER using RealmGui and standard SMuFL (Standard Music Font Layout) Bravura vector glyphs. Unlike external scoring applications, Notator runs directly inside REAPER's audio engine and timeline, providing instantaneous two-way synchronization between visual music notation and underlying REAPER MIDI takes, CC automation, and tempo maps.

1.2 The Three-Column Workspace Layout

The user interface is designed around an ergonomic three-column layout:

- Left Column (Sidebar Palette): Quick tools for note entry modes (pointer, pencil, text, eraser), rhythmic durations (whole to 32nd notes), augmentation dots, accidentals, tuplets, stem orientation, and articulation toggles.
- Center Column (Score Canvas): Infinite interactive notation canvas displaying visual staves, barlines, noteheads, stems, beams, ties, slurs, lyrics, and rehearsal marks.
- Right Column (Context Drawers & Tool Panels): Collapsible slide-out panels for Dynamics automation, Rearticulate banks, Tempo maps, Clef palettes, Key signatures, Pattern browser, and Settings.

1.3 Window Resizing & Docking

REAPER-Notator can run as a floating window or docked directly into any REAPER docker (bottom, top, left, right, or multi-tabbed). The score canvas dynamically scales according to the zoom factor (adjustable from 0.6x to 2.0x), preserving crisp vector typography at all DPI scaling factors.

Chapter 2: Top Toolbar & Global Navigation

2.1 Title & Version Indicator

Located in the upper left corner, displaying 'REAPER Notator' alongside the active build and semantic version number (v1.5.4). When running in a developer environment, Notator automatically detects and displays the active Git commit hash.

2.2 Transport & Playhead Controls

Functions:

- Rewind (|<): Instantly returns REAPER's edit cursor to measure 1.0 (time 0.0s).
- Play / Pause (> / ||): Starts timeline playback or pauses at the current cursor position. Key shortcut: Space.
- Stop: Halts playback and resets playhead according to REAPER's project settings.

DAW Effect: Calls REAPER's CSurf_OnPlay, CSurf_OnStop, and SetEditCurPos directly, maintaining sample-accurate alignment with REAPER's master audio engine.

2.3 Timeline & Meter Displays

Functions:

- Measure.Beat Display: Real-time read-out of playhead position in musical bars, beats, and quarter-note ticks (e.g. '004.01.00').
- Time Signature Indicator: Active project meter (e.g. 4/4, 3/4, 6/8, 7/8).
- BPM Indicator: Project master tempo at the current playhead position.

DAW Effect: Evaluates TimeMap_GetTimeSigAtTime on every frame to mirror timeline accelerandos, ritardandos, and time signature changes.

2.4 Audition Preview (Note Audio on Click)

Functions:

- Single-Note Audition: Clicking or vertically dragging any note on the canvas triggers an immediate acoustic preview sound, arbitrated to audition strictly the single hovered note even in dense chords or adjacent systems.
- Track Audition Isolation: Auditioning automatically isolates the target track and temporarily disarms concurrent project tracks, preventing Virtual MIDI Keyboard crosstalk across unrelated instruments sharing the same pitch/register.
- Articulation Chasing: The engine scans the project timeline from measure 1 up to the clicked note position to chase active Program Changes and note-level articulation marks (Staccato, Marcato, Tenuto, Pizzicato, etc.), dispatching the authentic Rearticulate bank and patch.
- Zero-Latency First Click: Implements a 1-frame pre-switch mechanism for Bank/Program Change before Note-On, eliminating sampler group switching lag so notes sound cleanly on the very first click.

DAW Effect: Dispatches synchronized Bank Select, Program Change, and instantaneous MIDI Note-On/scheduled Note-Off events to the active track's virtual instrument synth via REAPER's Audio Preview API.

2.5 Follow Playhead (Auto-Scroll)

Function: When active, the score canvas scrolls horizontally in real time to keep the active playback cursor centered in the viewport.

2.6 View Mode Selector

Functions:

- Auto: Automatically detects instrument clef based on track naming heuristics and pitch registers.
- Treble Only: Forces single G-clef staff rendering.
- Bass Only: Forces single F-clef staff rendering.
- Grand Staff: Renders classic piano/harp grand staff (Treble upper, Bass lower) connected by curly brace.

2.7 Track Picker & Visibility Filters

Function: Opens a multi-track routing popover allowing the user to select which REAPER tracks are currently visible, focused, or edited on the score canvas.

2.8 Multi-Voice System & Auto-Voice Overlap Splitting

Functions:

- 16 Polyphonic Voices: Maps MIDI channels 0-15 to 16 distinct engraving colors with configurable ghost voice opacity (5% to 100%).
- Auto-Voice: Intelligently analyzes the entire track for polyphonic overlaps and distributes them to channels 1-16.
- Auto-Split on Selection: Splits selected chordal intervals across independent voices with automatic opposite stem orientations.

2.9 Settings Modal & Unified Dark Mode Inversion

Functions:

- Collapsible Categorization: Settings are structured into 5 collapsible category headers (Theme & Appearance, Engraving & Notation, Audition & Playback, Shortcuts, and Advanced).
- Single Vertical Scroll Container: The entire modal body scrolls smoothly within a single child window, preventing double scrollbars and layout clipping.
- Static Action Footer: Pinned at the bottom with quick access to Reset and Close actions.
- Real-Time Shortcut Filter: Interactive search bar to quickly locate and rebind key actions.
- Comprehensive Dark Mode Inversion: Full palette inversion across all score elements, including noteheads, stems, beams, clefs, time signatures, rests, ties, lyrics, rehearsal marks, fermatas, and polyphonic voice colors.

Chapter 3: Score Canvas & Note Editing Engine

3.1 Note Entry & Pitch Snapping

Functions:

- Click on Staff: Inserts a new note with the duration selected in the sidebar at the nearest diatonic line or space.
- Ledger Lines: When moving beyond the 5 staff lines, Notator automatically calculates and renders standard-compliant ledger lines above or below the staff.

DAW Effect: Inserts a new MIDI note event into the focused track's active MIDI take via REAPER's MIDI_InsertNote API.

3.2 Note Duration & Augmentation Dots

Functions:

- Note Values: Supports 1/1 (Whole), 1/2 (Half), 1/4 (Quarter), 1/8 (Eighth), 1/16 (16th), and 1/32 (32nd) notes.
- Augmentation Dot: Toggles 1.5x duration expansion. A dotted quarter note spans 1.5 quarter notes (3 eighths).

DAW Effect: Sets note start and end positions accurately in Quarter Note (QN) timeline units.

3.3 Accidentals (Sharps, Flats, Naturals)

Functions:

- Flat (b): Decreases chromatic pitch by 1 semitone (-1).
- Natural (nat): Cancels preceding sharp/flat accidentals according to Western music engraving conventions (0).
- Sharp (#): Increases chromatic pitch by 1 semitone (+1).

DAW Effect: Transposes the underlying MIDI note number (0-127) and tags visual notation accidentals.

3.4 Note Manipulation (Move, Pitch, Length)

Functions:

- Horizontal Drag: Moves notes forward or backward in time, snapping to the selected rhythmic grid (e.g. 1/4, 1/8, 1/16).
- Vertical Drag: Transposes note pitch chromatically. Moving noteheads automatically updates accidentals.
- End-Edge Drag: Lengthens or shortens note duration by pulling the right edge of the notehead.
- Arrow Keys: Up/Down transposes by semitones (Shift+Up/Down by octaves). Left/Right moves by grid increments.

DAW Effect: Calls MIDI_SetNote to update pitch, QN start, and QN end in real time with undo history.

3.5 Multi-Note Selection & Marquee Tool

Functions:

- Marquee / Box Selection: Click and drag on empty canvas space to draw a selection rectangle encompassing multiple notes across measures and staves.
- Shift + Click: Add individual notes to selection.
- Delete / Backspace: Deletes all currently selected notes simultaneously.

DAW Effect: Multi-note batch deletion and batch transposition with atomic REAPER undo block.

3.6 Second-Interval Collision & Voice Separation

Functions:

- Second Intervals (Seconds): Notes placed on adjacent staff degrees in the same chord automatically offset horizontally (left/right) according to Elaine Gould engraving rules to prevent notehead collisions.
- Multi-Voice Stems: Polyphonic voices automatically flip stems (Voice 1 stems up, Voice 2 stems down).

3.7 Standard Gould Rests

Functions:

- Empty Measure Rests: Empty measures automatically display a centered whole-measure rest glyph.
- Rhythmic Rest Decomposition: Gaps between notes decompose into standard rests (quarter, eighth, 16th rests) strictly aligned to the meter division.

3.8 Non-Mouse Keyboard Note Navigation & Selection Expansion

Functions:

- Alt + Left / Right Arrows: Sequential note navigation along the timeline without mouse usage, selecting the previous or next note.
- Alt + Shift + Left / Right Arrows: Extends and expands note selections chronologically across the measure and score.
- Auto-Scroll Synchronization: When Follow Playhead / Auto-Scroll is active, the score canvas smoothly glides to keep newly selected notes centered in view.

Chapter 4: Standard-Compliant Automatic Beaming Engine

4.1 Meter-Based Beat Grouping

Functions: Consecutive eighth, 16th, and 32nd notes automatically connect with solid beams. Beaming boundaries strictly observe meter division (e.g. 4/4 groups into two half-measure halves; 6/8 groups into two dotted-quarter pulses of three eighths each).

4.2 Slope Calculation & Melodic Contour

Functions: Beams calculate a graceful visual slant following the pitch direction of noteheads. Extreme melodic intervals apply Gould slant-clamping to prevent excessive angles, and flat passages render strictly horizontal beams.

4.3 Fractional Beams (Beamlets & Stubs)

Functions: Syncopated rhythms and mixed subdivisions (such as a dotted eighth followed by a 16th note) generate standard fractional beamlets (stubs) oriented toward the rhythmic pulse beat.

4.4 Stem Inversion ('X' Shortcut)

Functions: Pressing the 'X' key instantly flips the stem direction of selected notes (stems up vs. stems down). Inverting stems automatically recalculates beam anchor points and flag alignments.

4.5 Hardware-Accelerated Beaming & Smooth Edge Anti-Aliasing

Functions:

- Clockwise Vertex Winding: All beam quads (primary, secondary, and fractional beamlets) enforce strictly clockwise vertex ordering in Dear ImGui screen space (top-left -> top-right -> bottom-right -> bottom-left). This guarantees outward-facing anti-aliasing normals with silky smooth, non-jagged edges across both upward and downward stem directions.
- Outer Stem Coverage: Beam polygons extend by half stem thickness ($1.25 * s$) on outer stems to encompass outer stems completely without horizontal protrusion (per Gardner Read & Elaine Gould).
- Embedded Stem Terminations: Stem lines terminate slightly inside the beam thickness to ensure flat rectangular line caps remain invisible within the slanted beam polygon.
- Gould Quarter-Note Quintuplets (5:4): Full 4/4 bar quintuplets accurately calculate 0.8 QN step widths spanning exactly 4.0 QN total duration.

Performance: Minimizes C-API overhead and eliminates redundant outline strokes.

Chapter 5: Dynamics Engine, Hairpins & Bow Swell Function

5.1 Standard Dynamic Levels (ppp to fff)

Functions: The Dynamics Drawer features 10 one-click dynamic badges:

- ppp (Pianississimo) -> Velocity ~20, CC ~25
- pp (Pianissimo) -> Velocity ~35, CC ~40
- p (Piano) -> Velocity ~50, CC ~55
- mp (Mezzo-piano) -> Velocity ~65, CC ~68
- mf (Mezzo-forte) -> Velocity ~80, CC ~82
- f (Forte) -> Velocity ~95, CC ~98
- ff (Fortissimo) -> Velocity ~110, CC ~112
- fff (Fortississimo) -> Velocity ~125, CC ~127
- sfz (Sforzando) / fp (Forte-piano): Sudden accent followed by immediate decay.

DAW Effect: Inserts dynamic text markings on the score and scales note velocities and continuous controller points.

5.2 Multi-Target CC Automation

Functions: Dynamics can write continuous automation curves to:

- CC1 (Modulation Wheel) - Standard for orchestral dynamics in cinematic sample libraries.
- CC11 (Expression) - Secondary loudness / timbre controller.
- CC7 (Main Volume) - Master channel volume.
- Velocity Only - Traditional keyboard velocity scaling.

DAW Effect: Generates dense, sample-accurate MIDI CC curves in REAPER's MIDI take envelope.

5.3 Hairpins (Crescendo & Diminuendo)

Functions:

- Crescendo (<): Visual opening wedge representing gradual increase in loudness.
- Diminuendo (>): Visual closing wedge representing gradual decrease in loudness.
- Dual Drag Handles: Circular handles at the start and end of hairpins allow exact quarter-note positioning.
- Curvature Selection: Toggle between linear ramps and exponential curves for organic acoustic swelling.

DAW Effect: Inscribes smooth CC ramps between the bounding dynamic levels.

5.4 Bow Swell Function & Bow Position Slider

Functions:

- Bow Swell Mode: Simulates acoustic string and brass swelling where a single sustained note or phrase swells up to a climax and decays back down.
- Bow Position Slider (0.0 to 1.0, default 0.5): Configures the exact peak inflection point of the swell per MIDI item. Setting 0.5 places the peak in the exact center; setting 0.8 creates an expressive late swell; setting 0.2 creates an explosive early swell.
- Per-Item Tuning: Bow swell parameters can be customized individually per MIDI item.

DAW Effect: Calculates an asymmetric Bezier CC curve mapped directly into REAPER's CC lane.

5.5 Phrasing & Shaping Bypass

Function: Checkbox to temporarily disable CC automation playback without deleting visual dynamic markings on the score canvas.

Chapter 6: Articulations & Rearticulate Integration

6.1 Rearticulate Bank Auto-Discovery

Functions: REAPER-Notator automatically scans REAPER's user directory for Rearticulate sound bank definitions (Rearticulate.reabank). When a track is selected, Notator matches track names (e.g. 'Violin I', 'Cello', 'Horns') to corresponding library banks.

6.2 Visual Playing Technique Glyphs

Functions: Provides an instant-access drawer for score playing techniques:

- Staccato (dot), Staccatissimo (wedge), Accent (>), Marcato (^), Tenuto (-)
- Pizzicato (pizz.), Arco, Con Sordino, Sul Ponticello, Col Legno, Tremolo, Harmonics (o)

Placement: Notator automatically positions articulation marks above noteheads for stems-down notes, or below noteheads for stems-up notes according to standard engraving conventions.

6.3 DAW Automation Impact

DAW Effect: Inserting an articulation writes:

1. REAPER Type 15 notation text events into the MIDI take for persistent score recall.
2. MIDI CC0 / CC32 Bank Select and Program Change messages at the note start position to trigger sample library key switches in Kontakt, Spitfire, VSL, Orchestral Tools, etc.

Chapter 7: Holding & Sustain Pedal Lane (CC64)

7.1 Dedicated Pedal Lane

Functions: Renders continuous piano sustain pedal markings positioned below the bass staff with automatic collision clearance from low notes and dynamic hairpins.

7.2 Three Historical Engraving Styles

Functions:

- Classic: Ped. symbol at start, dashed horizontal line, and asterisk (*) at release.
- Bracket: Modern square brackets ([---]) with vertical hooks.
- Notch / Mixed: Combines Ped. marking with inverted 'V' notches for continuous pedal retakes.

7.3 Dual Handles & Pause / Break Retakes

Functions:

- Start & End Handles: Drag circular handles to adjust pedal engage and release times.
- Pause / Retake Points: Right-click on the pedal line to insert pedal breaks (quick release and re-engage) without creating multiple separate items.

DAW Effect: Writes CC64 value 127 at engage, momentary 0 at retakes, and 0 at final release.

Chapter 8: Tempo Maps & Metric Transitions

8.1 Absolute Tempo Markers

Functions: Insert tempo markers with custom BPM values (e.g. Quarter = 120, Dotted Quarter = 72) and descriptive Italian tempo text (Adagio, Andante, Allegro, Presto).

DAW Effect: Inserts a master tempo marker directly into REAPER's timeline tempo envelope.

8.2 Gradual Transitions (Accelerando / Ritardando)

Functions: Renders dashed tempo transition lines spanning multiple measures (e.g. 'poco a poco accel. -----').

DAW Effect: Generates a continuous gradual tempo ramp in REAPER, smoothly accelerating or decelerating project playback speed.

8.3 Direct Double-Click BPM Editing & Glyphs

Functions:

- Double-Click Inline Editing: Double-clicking any tempo marker opens an instantaneous popup editor with automatic keyboard focus on the numeric BPM input.
- Standard Unicode Note Symbol: Renders the classical quarter-note symbol ([Quarter], U+2669) in non-bold regular weight according to professional music engraving standards.

DAW Effect: Updates REAPER tempo markers in real time with immediate timeline synchronization.

Chapter 9: Octave Shift Lines (8va, 8vb, 15ma, 15mb)

9.1 Visual Score Simplification

Functions: For extreme high or low passages that would otherwise require excessive ledger lines, octave shift lines simplify score reading:

- 8va (Ottava Alta): Notes sound 1 octave higher than written.
- 8vb (Ottava Bassa): Notes sound 1 octave lower than written.
- 15ma (Quindicesima Alta): Notes sound 2 octaves higher.
- 15mb (Quindicesima Bassa): Notes sound 2 octaves lower.

9.2 Visual vs. Sounding Pitch

DAW Effect: Notes remain in their natural visual position on the staff for effortless reading, while underlying MIDI note pitches trigger the intended high or low acoustic octaves.

Chapter 10: Orchestral Pattern Browser & 1-Click Library

10.1 Curated 1,200 Pattern Factory Library

Functions: Slide-out pattern browser organized into 8 orchestral categories (1,200 factory patterns total):

- 1. Strings Staccato (150x driving cinema pulses, ostinatos)
- 2. Strings Pizzicato (150x delicate plucks, agile grooves)
- 3. Brass Blockbuster (150x epic horn fanfares, low brass power)
- 4. Cinematic Melodies (150x soaring lyrical themes)
- 5. Counter Melodies (150x rich orchestral counterpoint)
- 6. Woodwinds Textures (150x runs, fluttering textures)
- 7. Cinematic Piano (150x flowing arpeggios, grand staff ballads)
- 8. Ancient Greek & Roman Harp (150x modal hymns, Delphic paeans, Sapphic strophes, Dorian/Phrygian/Lydian processions)
- 9. Custom User Patterns (user-captured motifs and items)

10.2 One-Click Insertion, Drag & Drop & Audio Audition

Functions:

- Drag & Drop: Drag any pattern tile directly onto the score canvas with golden ghost-preview.
- Audio Audition: Audition motifs in real-time through any selected REAPER instrument track.
- Capture Selection: Select any REAPER MIDI item and click 'Capture REAPER Item' to store it permanently.

10.3 In-App 1-Click Factory Library Downloader

Functions: Integrated one-click downloader built into the browser toolbar and empty-state banner.

- 1-Click Install: Downloads and unpacks the entire 1,200 pattern library (820 KB compressed) in one second without leaving REAPER.
- Offline Archive Support: Automatically unpacks local patterns.zip on first launch if present.

Chapter 11: Rehearsal Marks, Fermatas & Score Tools

11.1 Dynamic Rehearsal Marks & Interactive Drag-and-Drop

Functions: Professional structural navigation and rehearsal tagging:

- Dedicated Rehearsal Lane: Positioned cleanly between the Chord Track lane and Bar Numbers, providing unobstructed structural visibility across all systems.
- Interactive Mouse Drag & Drop: Click and drag any rehearsal mark directly along the top lane to move it to any measure. During dragging, the mark follows the cursor with a real-time vertical snap guide line and target bar indicator ('Bar X').
- Auto-Sequencing: Marks automatically sequence as [A], [B], [C]... or [1], [2], [3]... Moving or inserting marks chronologically re-indexes all subsequent marks across the score.
- Edit-Cursor Placement: The Tools Drawer dynamically detects the active REAPER edit cursor position, displaying explicit actions like 'Add Letter Mark at Bar X' and 'Add Number Mark at Bar X'.
- Classical Navigation Symbols: Immediate one-click placement of Da Capo (D.C.), D.C. al Fine, Dal Segno (D.S.), D.S. al Coda, Segno, Coda, and Fine marks.
- Inverted Engraving Typography: Styled according to classical engraving rules with a crisp paper background, 2px dark border, and high-contrast dark typography.
- Settings & View Options: Configurable vertical Y-offset slider in Settings and a show/hide toggle in the View dropdown.

11.2 Score-Wide Vertical Fermatas

Functions: Classical pause and hold articulation across all staves:

- Four Engraving Types: Standard fermata, Short fermata (triangle/fermata corta), Long fermata (square/fermata lunga), and Very Long fermata.
- Full-Score Clickability: Fermatas can be clicked, selected, and edited on any staff across the entire vertical score system, not just the top staff.
- Tempomap Playback Coupling: Non-destructive tempo slowdown dip (1.25x to 3.0x multiplier) with automatic restoration at the release point.
- Keyboard Delete Support: Select any fermata or rehearsal mark and press Delete or Backspace to instantly remove it.

11.3 Score Tools Drawer & Performance Transformations

Functions: Fast editing actions consolidated in the right-hand Tools drawer:

- Make Notes Legato: Extends note durations to adjacent note downbeats for seamless cantabile phrasing.
- Auto Voice & Auto Voice on Selection: Polyphonic splitting of selected chords into upper Voice 1 (stems up) and lower Voice 2 (stems down).
- Arpeggio Strums: Realistic harp and guitar roll simulation with upward/downward wavy line engraving and non-destructive playback micro-offset.
- Quantize Tools: Integrated triplet, swing, and 1/4 through 1/64 grid snapping directly inside the Tools drawer.

11.4 Floating Text Items & In-Place Editing

Functions: Pressing Ctrl+T inserts a floating score text item at the cursor. Double-click any text item to open an in-place editing field. Configure font size, standard/italic/bold styling, and staff attachment anchor points.

Chapter 12: MusicXML 4.0 Import & Export

12.1 Interoperability Standard

Functions: MusicXML 4.0 is the universal interchange format between professional notation software. REAPER-Notator includes a dedicated MusicXML parser and exporter written in pure Lua.

12.2 Importing Scores & Articulation Auto-Mapping

Functions:

- **Orchestral Score Import:** Click 'Import MusicXML' to load orchestral scores created in external notation programs. Notator parses parts, measures, time signatures, key signatures, pitch data, dynamics, lyrics, and tempo markers, creating fully arranged tracks and MIDI items directly in REAPER.
- **Intelligent Articulation Detection:** Automatically analyzes Reaticulate sound banks loaded on destination tracks. Employs a prioritized 3-tier mapping algorithm for articulations (e.g. prioritizing dedicated 'Staccato Dig' patches for Spitfire Double Bases over fallback 'Short 0.5' duration patches).
- **Compact Track Layout:** Automatically collapses imported track heights (25px) to provide an immediate, organized orchestral overview without overwhelming the REAPER track arrangement.

12.3 Exporting Scores

Functions: Click 'Export MusicXML' to save your REAPER project as a standard .musicxml file ready for publication, live orchestral recording sessions, or further engraving.

Chapter 13: Score Clipboard & Selective Copy/Paste Engine

13.1 Selective Note & Score Clipboard

Functions: High-precision clipboard operations (Ctrl+C / Ctrl+V) with strict element isolation:

- Selective Note Copying: When copying selected notes, the clipboard selectively captures notes, chords, and explicit note articulations (staccato, accent, tenuto, fermatas) without inadvertently dragging along unselected dynamics, hairpins, pedal lines, or tempo markers.
- Context-Aware Pasting: Pasting notes into a MIDI item respects the existing destination item boundaries and timeline cursor position without truncating or unexpectedly expanding underlying media items.
- Independent Element Duplication: Dynamics, hairpins, and pedal lines can also be copied and pasted independently, ensuring modular workflow efficiency.

13.2 Quantization & Sub-Tick Boundary Alignment

Functions: Precision alignment engine across copy/paste and editing routines:

- PPQ Temporal Integrity: Pasted notes maintain exact sub-tick delta relationships relative to the edit cursor.
- Track-Targeting: Clipboard contents paste directly into the active or focused track, enabling rapid passage duplication across orchestral sections.

Chapter 14: Engine Architecture & Performance Optimization

14.1 High-Efficiency O(1) Project State Caching

Functions:

- Project State Guard: Monitors `reaper.GetProjectStateChangeCount(0)` to eliminate thousands of redundant C-API queries during playback and idle, reducing C-API calls to 0 when the score is static.
- Text Size Memoization: Caches ImGui font measurement dimensions to eliminate repeated layout calculation storms.
- RealmGui Context Validation: Strict pointer validation guarantees rock-solid stability during project switching and window operations.

14.2 Hardware-Accelerated Vector Graphics

Functions:

- Pure GPU-Rasterized Drawing: RealmGui dispatches all vector paths, noteheads, and beams directly to DirectX 11 / OpenGL / Vulkan.
- Optimized Beam Meshing: Eliminated redundant contour stroking on filled polygons, reducing vertex count by 60% and halving C-API call overhead.

Chapter 15: Keyboard Shortcuts & Quick Reference

15.1 Comprehensive Hotkey Matrix

Quick reference table for high-speed score entry:

Key / Shortcut	Action	Description
:---	:---	:---
Space	Play / Pause	Starts or stops timeline playback
Delete / Backspace	Delete	Deletes selected notes, dynamics, tempo marks, or lines
Esc	Clear Selection	Deselects all active items and clears focus
Up / Down Arrows	Transpose Pitch	Moves selected notes up/down by 1 semitone
Shift + Up / Down	Octave Shift	Moves selected notes up/down by 1 octave (12 semitones)
Left / Right Arrows	Nudge Position	Shifts notes horizontally by the active grid increment
Shift + Left / Right	Resize Duration	Shortens or lengthens selected notes by grid step
Alt + Left / Right	Select Note	Navigates to previous / next note on timeline without mouse usage
Alt + Shift + Left / Right	Expand Selection	Expands multi-note selection along the timeline
X	Invert Stem	Flips note stem direction (up <-> down)
1 .. 6	Rhythmic Values	1=Whole, 2=Half, 3=Quarter, 4=Eighth, 5=16th, 6=32nd
. (Period)	Toggle Dot	Toggles dotted note duration (1.5x)
- / 0 / =	Accidentals	-=Flat (b), 0=Natural (nat), ==Sharp (#)
T	Tie Note	Toggles tie / Bindebogen to adjacent note
S	Slur Phrase	Toggles legato slur over selected passage
A	Accent	Toggles accent mark (>) on selected note
Ctrl + T	New Text Item	Creates a floating text annotation at cursor position
Ctrl + S	Save Project	Triggers REAPER project save